Roll Number:_____

Thapar Institute of Engineering & Technology

Department of Computer Science and Engineering

END SEMESTER EXAMINATION

B. E. (2nd Yr. COE/CSE)
19th May, 2026
Time: 9:00 am to 12:00 pm
Max Marks: 60 (Weightage: 40)

Course Code: UCS310
Course Name: Database Management System
Name of Faculty: Dr. Simranjit (C6), Dr. Shashank (C3), Dr. Amrita (C1), Dr. Mahak (Q1, Q2), Dr. Shivani (C8, C5, Q4), Dr. Ananya (Q3, C7), Dr. Tanya (C2, C4), Dr. Manisha (P1)

Note:Attempt any four Questions in sequence. Answer all sub-parts of each question at one place. Do mention Page No. of your attempt at front page of your answer sheet. Assume missing data (if any).

Q.No	Questions	MM	CO	BL
Q.1	A nationalized bank is developing a Sustainable Digital Banking System to promote secure, transparent, and technology-driven financial services. The bank maintains customer account information in the following relation: **account (acc_no, customer_name, branch_name, balance)** As a part of the software team, you need to implement following PL/SQL programs for balance management, account monitoring and secure fund transfers. (i) Write a PL/SQL block to identify the account having the minimum balance in branch 'Patiala' and increase the balance by ₹5000. (ii) Create a PL/SQL function check_balance_status(acc_no) that returns 'Low Balance' if balance is below ₹10000, otherwise returns 'Sufficient Balance'. (iii) Write a PL/SQL procedure for a banking application that transfers funds from one account to another. The procedure must ensure that the sender maintains a minimum balance of ₹1000 after the transaction. If the balance condition is violated, an appropriate exception should be raised, and the entire transaction must be rolled back.	(5) (5) (5)	CO5	L3

Q.2	**a)** Explain the ACID properties of transaction management.
	b) Consider the following schedule involving three transactions T1, T2, and T3 operating on data items A and B:
	S: $R_1(A), W_2(A), W_1(B), R_3(B), R_2(B), W_3(A)$
	List all conflicting operations and construct the precedence graph for the above schedule. Determine whether the schedule is conflict serializable or not.
	c) Consider the following two transactions:
	Transaction T1 / **Transaction T2**
	read(X); / read(Y);
	read(Y); / read(X);
	if X > 50 then / if Y < 100 then
	Y := Y + 10; / X := X - 20;
	write(Y); / write(X);
	Add appropriate lock and unlock instructions so that both transactions follow the Two-Phase Locking Protocol. Determine whether the execution of these transactions can lead to a deadlock.
	d) A database system takes a checkpoint while transactions T1 and T2 are active. After the checkpoint, T1 commits but T2 does not, and the system crashes. Explain how checkpoint helps in reducing the recovery work in this case.
		(4)
		(4)
		(5)
		(2)
			CO4	L1
				L4
				L3
				L2

Q.3	**a)** A university is developing a Smart Academic Management System to improve the quality, accessibility, and organization of higher education services. Following relation is maintained by a university: **Student_Course (S_ID, C_ID, ID I_R, G)** where S_ID, C_ID, I_ID, I_R and G stands for student id, course id, instructor id, instructor room and grade respectively. Following functional dependencies are given:	(7)	CO3	L4

	S_ID, C_ID → I_ID, G I_ID → I_R As a part of the university database team, your task is to analyze and normalize the relation to improve database efficiency, eliminate redundancy, and ensure consistency in academic records. (i) Find the candidate key(s) of the relation. (ii) Determine whether the relation is in 2NF with proper justification. (iii) Determine whether the relation is in 3NF. If not, decompose the relation into 3NF. (iv) Introduce the multivalued dependency in the relation by adding one attribute and justify your answer. **b) State whether each of following statements is True/False. Justify your answer.** (i) A relation having only trivial functional dependencies is automatically in BCNF. (ii) Denormalization may increase the chances of update anomalies in a database system. (iii) If a relation has only one candidate key consisting of a single attribute, then it violates 2NF. (iv) If every functional dependency in a relation has a single attribute on RHS, then the relation is in 3NF.	(8)		L2
Q.4	**a)** A library maintains records of Books and Members. Each book has a unique BookID, a Title, a Genre, and a Price. The PublishYear of a book is stored, and the Age of the book can be calculated from it. Each member has a unique MemberID, a Name (made up of FirstName and LastName), and can register more than one phone number. A member can borrow multiple books and multiple copies of a book are available in the library. Each borrowing transaction has a BorrowDate recorded with it. (i) Draw the complete ER diagram for the above description. (ii) Convert the ER diagram into a relational schema.	(8)	CO2	L3
	b) A national poverty eradication program maintains records of rural families, government welfare schemes, NGOs, and financial assistance programs. Initially, the system stored data using a Hierarchical data model. Later, the organization considered the Network data model due to increasing complexity. Finally, the government migrated the system to a Relational data model. Explain how data would be organized in the three data models for this poverty management system. Also, compare the Hierarchical and Relational model in terms of redundancy and data independence.	(7)	CO1	L2
Q5	**a)** A municipal corporation maintains a water consumption database to provide clean water and sanitation. A PL/SQL cursor scans all records in ascending order of HouseID. If MonthlyConsumption exceeds 20,000 litres, WaterBill is increased by 15%. Otherwise, WaterBill is increased by 5%. A trigger automatically changes the Status to: "Excessive Usage Warning" whenever the updated WaterBill exceeds ₹4000.	(7)	CO5	L3

HouseID	FamilyName	MonthlyConsumption (Litres)	WaterBill	Status
H101	Sharma	12000	1500	Normal
H102	Kaur	18000	2200	Normal
H103	Ali	25000	3000	Normal
H104	Verma	32000	4200	Normal

(i) Determine the final **WATER_USAGE** table after execution of the cursor and trigger. Show all calculations clearly.
(ii) What are advantages of using triggers in smart water management systems.

b) Consider the following log sequence of two transactions T1 and T2 operating on a bank database. The initial balance of account B = 12000 and mortgage account M = 0. (8) CO4 L4

1. T1 start
2. T1: B old = 12000, new = 10000
3. T1: M old = 0, new = 2000
4. T1 commit
5. T2 start
6. T2: B old = 10000, new = 10500
7. T2 commit

Assume that the database system crashes just before **log record 7** is written to stable storage.
(i) Identify which transactions are committed and uncommitted at the time of crash.
(ii) Explain the recovery procedure to be applied during system restart. Clearly specify which log records will be undone/redone. Also, determine the final values of accounts B and M after recovery.
(iii) Justify why redo is not applied to uncommitted transactions and why undo is necessary.